

Taran Govindu

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EDUCATION

Georgia Institute of Technology

Bachelor of Science in Aerospace Engineering

Expected Graduation May 2028

4.0 GPA

- Relevant Courses: Statics, Engineering Materials, MATLAB, CAD, Physics (E&M, Mechanics)

EXPERIENCE

Propulsive Landers at Georgia Tech

Atlanta, GA

Developing a throttleable hybrid rocket (N_2O /paraffin) with thrust vector control for autonomous propulsive landing
Propellant Feed Systems Engineer Jan 2026 – Present

- Designed a servo-actuated ball valve assembly in SolidWorks to replace a ~5 lb electric motor actuator, reducing valve actuation mass to ~0.7 lb while maintaining flow regulation requirements within a 5.5 lb PFS mass budget
- Sized regulators, burst disks, and valves across the propulsion feed system by deriving pressure and flow requirements from first principles, evaluating 10+ candidate components per subsystem against mass and performance constraints, and produced a final bill of materials that has been ordered and is entering integration
- Leading the ground fill system from concept through implementation: developed the P&ID for filling nitrogen and nitrous oxidizer tanks from an off-vehicle source, sized components, and am selecting parts under operational and safety constraints while documenting design rationale for each decision

GNC Simulation Engineer

Jan 2026 – Present

- Building a 6-DOF flight dynamics simulation in Rust with stochastic wind disturbances, aerodynamic lookup tables, and non-centered moment resolution to serve as a digital twin for testing thrust vector control algorithms against vehicle trajectory response
- Integrating subsystem models including valve actuation dynamics, pressure transducer response, and RCS valve sequencing to capture how propulsion feed system behavior propagates into vehicle-level flight dynamics

Exascale CFD VIP at Georgia Tech

Atlanta, GA

Researcher

Jan 2026 – Present

- Engineered a multi-scheme Burgers' equation solver in C++ supporting Godunov and Lax-Wendroff methods with a modular architecture for integrating additional numerical schemes into a unified executable
- Simulated canonical flow phenomena (Gaussian pulse, N-wave) across viscosity regimes to validate solver convergence and evaluate numerical dissipation between schemes; extended the Godunov solver from 1D to 2D

PROJECTS

Verus | Autonomous Bridge Inspection Platform

Dec 2025 – Present

- Engineering an autonomous inspection drone in SolidWorks with GD&T and DFM considerations, designing the airframe to package purchased electronics and sensor hardware while validating path-planning algorithms in a Python/PX4/Gazebo testbed prior to hardware integration
- Built a CNN-based classifier in Python to detect subsurface delamination in bridge decks from ground-penetrating radar data, and developed a synthetic data generation pipeline using gprMax FDTD simulations to produce ~50,000 labeled training signals, addressing real-world data scarcity
- Secured \$150,000 in funding (valuing company at \$1.5M) from Fusen World and secured \$5,000 from Georgia Tech's Startup Incubator (Create-X) for technical viability and market potential

Acoustic Drone Detection System | 36-Hour Hackathon

Apr 2026

- Reverse-engineered a speaker to function as a microphone, designed a parabolic dish in CAD to amplify noise detection, soldered the sensors to LED indicators and digital display, and built full system costing less than \$25
- Performed live Fast Fourier Transforms on microcontroller to isolate drone propeller frequency bands from background noise, then trained neural network on 20,000+ audio samples in Python and deployed the trained model in C++ for on-device inference, achieving 85% detection accuracy on live audio with no internet dependency

SKILLS

Programming: Python, Rust, C++, MATLAB, Java, Bash

Simulation & CAD: SolidWorks (GD&T/DFM), CATIA, Fusion 360, Ansys (Fluent/Mechanical)

Robotics & Controls: PX4 Autopilot, ExpressLRS, 6-DOF Modeling, Gazebo, PID Control

Infrastructure & Tools: Soldering, 3D Printing, Multimeters, Linux (Ubuntu), PACE High Power Computing